

Part: **3.3 – METAL ASSEMBLIES**
Chapter: **3320**
Title: **CURVED STIFFENED PANELS**
Revision: **0.5**
Date: **27/05/2022**

1 INTRODUCTION

1.1 OBJECT

This document proposes a stress sizing method for the metallic skin and stringers of a cylindrical fuselage.

The method discussed in this chapter corresponds essentially to that proposed by Melcon and Ensrud in Ref. 7.

1.2 SCOPE AND LIMITATIONS

Scope and basic limitations:

- ✓ Panels made of metallic alloys with some degree of curvature (cylindrical shape) stiffened with longitudinal stringers in axial direction, effectively supported at frame stations
- ✓ Curvature radius of skin not higher than 2500mm
- ✓ Closed form (non-numerical) methods with the minimum necessary simplification assumptions and limited accuracy
- ✓ Each skin subpanel is modelled as a cylindrical shell, with uniform thickness and perimeter simply supported
- ✓ The structural element considered in this analysis is the super-stringer, composed by the stringer and adjacent subpanels (halves), effectively supported at frames stations

See details in following pages...

2 STRUCTURAL DESCRIPTION

The structural configuration analysed by this method, the super-stringer. This idealized element consist of the piece of stringer plus the (half) adjacent skin panels, limited at its ends by the frames supports.

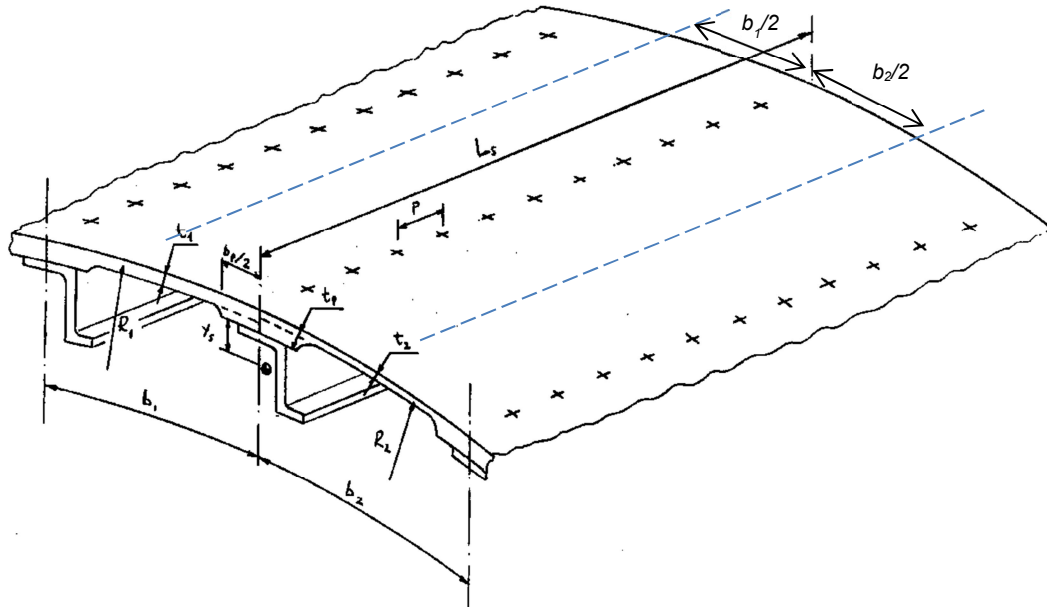


Figure 1. Sketch of curved stiffened panel between frames supports

Some remarks and assumptions in this structural model and method (see details in following pages):

- Axial loads are condensed in the super-stringer cross-section forcing compatibility of longitudinal strain across the super-stringer section. On the other hand, shear and transversal loads are taken specifically for each of the two skin adjacent sub-panels.
- When analysing a complete fuselage, each skin subpanel is typically checked twice (with slightly different results), as part of two different super-stringers
- The skin subpanels enclosed among frames and stringers ($L \times b$) are assumed simply supported at all its edges (additional torsion stiffness is conservatively ignored in local stability analyses).
- The super-stringer column is effectively supported at frames stations, with the rotation restriction controlled by means of the fixity coefficient C

3 GENERAL STRESS & STRENGTH ANALYSIS PROCESS

The flow diagram below describes roughly the strength analysis process of a super-stringer location. It starts by calculating the pure-allowables, i.e. the allowable loads and strengths corresponding to the possible failure modes in uni-axial loads (material strengths, local buckling and crippling loads, column buckling...). Later on, for each load case, the process continues by calculating the stress distribution within the different details (taking into account non-linear post-buckling effects), the corresponding failure indices and finally obtaining the minimum reserve factor (which may require additional iterations for responses that are non-linear with applied loads).

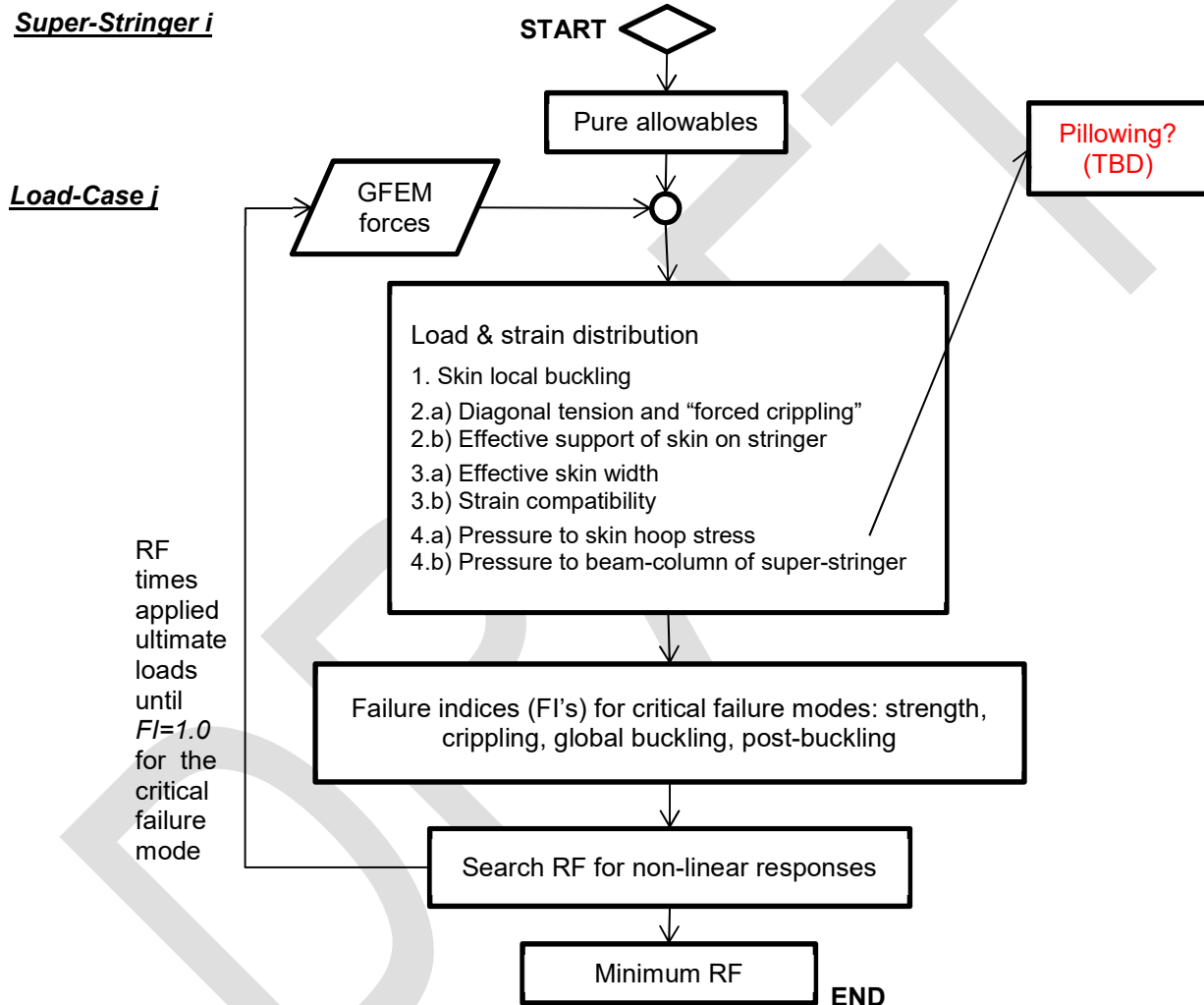


Figure 2. Flow Diagram of the standard strength check process of a super-stringer detail in a stiffened panel of an airframe fuselage

4 PURE ALLOWABLES

This sub-section discuss the allowable loads, beyond basic material strength, which can be derived from the structural detail independently of its complete stress status, corresponding to a set of failure modes which may occur under basic/uniaxial loads.

4.1 SKIN

Inter-rivet buckling

The inter-rivet buckling stress of the skin (pad) under the stringer attachment limits the compression longitudinal stress which can be locally sustained. It is used later on for calculating the effective skin width and the column buckling load of the super-stringer element.

The inter-rivet stress shall be calculated per method in chapter §3180 "INTER-RIVET BUCKLING".

Skin local buckling

The local buckling of the skin subpanels next to the stringer, and between frames supports, are needed for determining later the post-buckling behaviour of the structure. A (non-critical) margin related to local buckling of the skin shall be calculated and presented for information purposes (as post-buckling of skin below ultimate loads is typically permitted).

The buckling load under longitudinal compression shall be calculated per methods discussed in chapter §3150 "PLATE/SHELL BUCKLING". Note: legacy methods currently using, for this purpose, methodologies discussed in Ref. 1, Ref. 3, Ref. 4, Ref. 5.

For shear loads, however, in order to be consistent with the methodology used later to deal with post-buckling behaviour, the expressions proposed by Melcon (Ref. 7) shall be applied instead. Using the same notation in that reference, the allowable shear buckling stress of the curved panel shall be:

$$F_{scr} = F'_s + F''_s$$

where

$$F''_s = 5.25 \cdot E \cdot (t/b)^2$$

reference buckling shear stress of the flat panel

$$F'_s = [\pi / (4 \cdot S^{0.25})] \cdot (E \cdot t / R)$$

contribution of the curvature, with $S = a^2 / (R \cdot t)$

4.2 STRINGER

Stringer crippling

The crippling strength of the stringer under compression loads shall be calculated according to method defined in chapter §3130 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS".

Stringer column buckling

The column buckling load of the stringer, considered separated from the skin, due to bending instability around the axis parallel to the skin, shall be needed for later application of the post-buckling methodology proposed by Melcon (Ref. 7).

This buckling load shall be calculated according to method discussed in chapter §3140 "COLUMN BUCKLING IN COMPRESSION". The fixity coefficient recommended for stringers that are continuous across frames supports (both) is 2.0, which is equivalent to a column with guided/supported support conditions at its ends. This value should be reduced on designs with lower capability to retard buckling (eg. stringer run-out at/near any of its frames supports). Higher values of the fixity coefficient should be properly justified.

This allowable value, when expressed as stress shall be denoted by the F_c symbol.

4.3 SUPER-STRINGER (STRINGER PLUS ADJACENT PANELS)**Super-Stringer column buckling**

The column buckling load of the super-stringer assembly, with a section composed by the stringer and the (effective) skin width it is attached to, shall be also necessary for the later strength check.

Combined methodologies of chapter §3140 previously noted, §3185 "EFFECTIVE SKIN WIDTH" and §3180 "INTER-RIVET BUCKLING" are needed to compute this allowable load. A value typically recommended for the fixity coefficient in this analysis is 1.0 (equivalent to a column simply-supported at both ends). Greater values of this parameter could be used if the stringer/frame crossings are rigid enough and the factor is adequately justified.

5 LOADS AND STRESSES

5.1 INTERNAL (IN-PLANE) LOADS

The set of internal loads relevant for the structural analysis of this detail shall be

- Stringer axial/longitudinal load/force P_{xst}
- Axial and shear loads in 1st skin subpanel N_{x1} , N_{y1} , N_{xy1}
- Axial and shear loads in 2nd skin subpanel N_{x2} , N_{y2} , N_{xy2}

Then, the total axial load/force in the super-stringer cross-section is finally:

$$P_x = P_{xst} + 0.5 \cdot (b_1 \cdot N_{x1} + b_2 \cdot N_{x2})$$

Loads extraction from GFEM

Typical modelling of the fuselage skin and stringers, in a GFEM, is as follows.

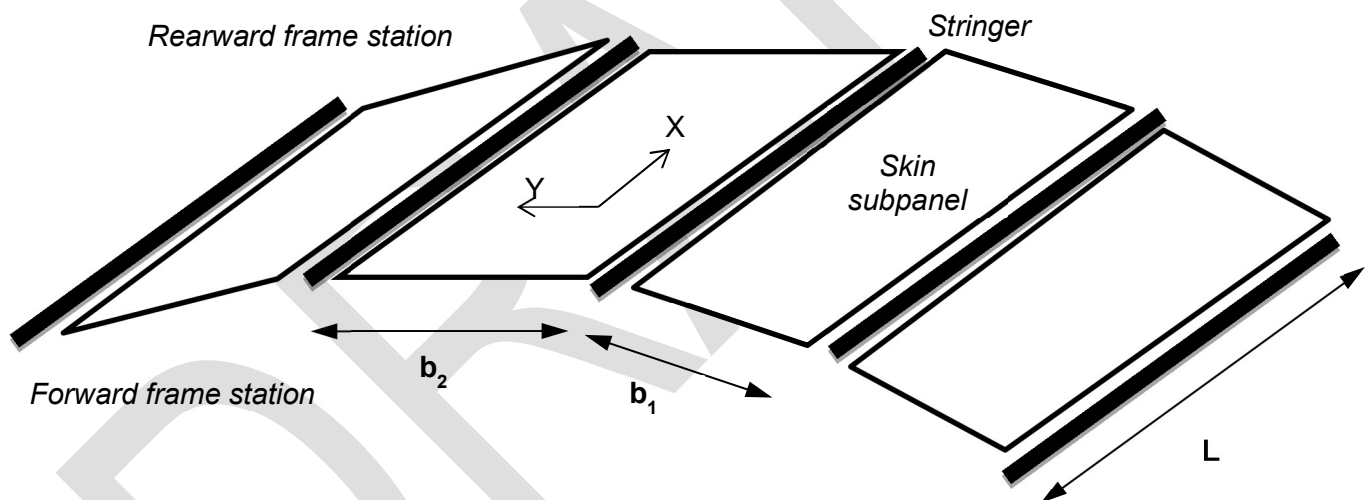


Figure 3. Stress model in GFEM

Notes:

- ✓ There may be more than one FEM element along the distance between support frames ("L" in figure above). In military transport aircrafts, usually there will be only one element between stringers. Normalization approaches of internal loads for later stress and buckling stability analyses shall typically include averaging and/or taking the elements with maximum and minimum forces.
- ✓ X axis in FEM elements local reference assumed parallel to longitudinal/stringer direction

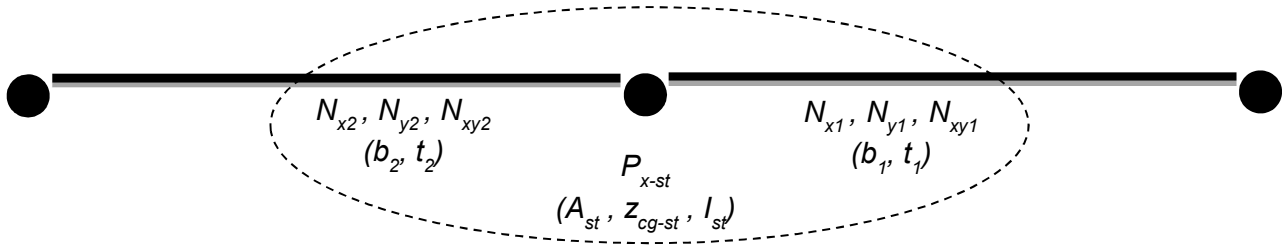


Figure 4. GFEM loads and areas of the super-stringer cross section at a certain location

5.2 LONGITUDINAL STRESS

Direct/primary compression stress in the super-stringer cross section, assuming materials of skin and stringer are the same, can be computed as:

$$f_{cg} = P_x / A$$

where A is the cross-sectional area of the stringer plus the halves of the skin adjacent subpanels.

In case materials are different, the different longitudinal stresses in stringer and skin shall be calculated imposing compatibility in axial stress and loads equilibrium.

5.3 SHEAR LOAD AND STRESS IN SKIN

Direct/primary shear stress in skin can be computed from skin shear load (flow) simply as:

$$f_s = N_{xy} / t$$

Diagonal tension

Instead of following the classical diagonal tension analysis proposed by Kuhn (Ref. 6), Melcon (Ref. 7) suggest a more direct empirical approximation consistent with his tests observations.

The longitudinal compression stress induced in the stringer by the diagonal tension stress field developed in the post-buckled skin in shear shall be:

- If $(f_s < F_{scr})$ then $f'_c = 0$
 - else $f'_c = \nu \cdot (f_s - F_{scr}) \cdot (b \cdot t / A_{st})$
- where $\nu = 1 + (a/R) \cdot [(I_{st}/J_{st}) \cdot (t/b)]^{0.25}$

Expressions for describing the stress status in the post-buckled skin shall not be shown here. Instead, for checking its strength, Melcon proposed a dedicated shear allowable stress with account for this effect (see §6.2).

Effective stringer inertia and global buckling

In order to assure the stringer is capable to prevent global buckling in shear (effectively separating skin in the subpanels limited by stringers), Melcon (Ref. 7) proposed a method in which this condition is checked by calculating a fictitious compression stress in the stringer which should be compared with the column buckling allowable stress of the stringer (separated from the skin). This fictitious stress is calculated as:

- If $(f_s < \lambda \cdot F'_s)$ then $f''_c = 0$
- else if $(\lambda \cdot F'_s \leq f_s < \lambda \cdot F_{scr})$ then $f''_c = [(f_s - \lambda \cdot F'_s) / F''_s] \cdot [\lambda \cdot F''_s \cdot a \cdot t / A_{st}] \cdot [(0.053 \cdot b/a)^{1/3}]$
- else if $(f_s \geq \lambda \cdot F_{scr})$ then $f''_c = [\lambda \cdot F''_s \cdot a \cdot t / A_{st}] \cdot [(0.053 \cdot b/a)^{1/3}]$

where

$$\lambda = [1 / (1 + f_{cg}/f_s)]^{0.25}$$

5.4 PRESSURE FORCES

The internal-loads calculation discussed in section §5.1 (and following) will not be sufficient if the curved panel is submitted to significant pressure forces (pressurized fuselages, fuel tanks in hard-landing cases...), especially if the source of internal loads is a GFEM and it does not explicitly use pressure forces applied onto the elements modelling the curved panels.

Hoop stress in skin

In cylindrical shells of curvature radius R , when subjected to (internal) pressure p , the natural response of the structure is to react this load as a uniform hoop stress in the shell. Thus, a delta transverse Y load ΔN_y shall be added to the panel.

$$\Delta N_y = p \cdot R \quad \dots \text{or, as hoop stress:} \quad \Delta \sigma_y = p \cdot R / t$$

This additional stress shall be taken into account when computing the safety margin of the skin panel submitted to direct loads (Von Mises point stress criterion, see §6.2). Nevertheless, this solution is not expected to be realistic near the supports of the panel (eg. near stringers or frames in fuselage skins).

Super-stringer as beam-column

The simplified approach to treat by analysis of super-stringer structure under pressure load shall be to integrate the pressure across the section (considering the area of half adjacent skin panels) and superimpose that stress status to that due to the in-plane forces discussed before (axial P_x force mainly).

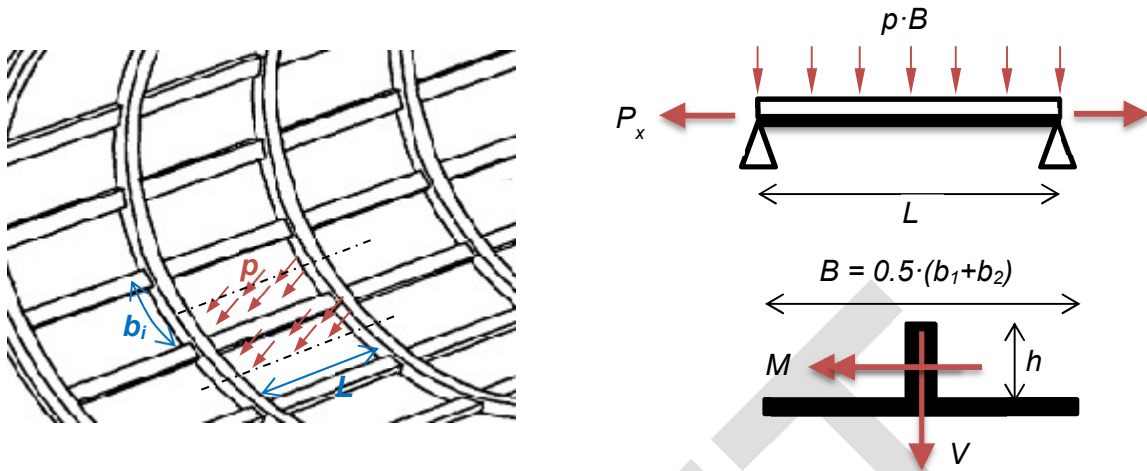


Figure 5. Super-stringer under (inner) pressure as beam-column

The internal loads in the super-stringer cross section, due to the combined effect of the direct axial force P_x and pressure on skin shall be:

- Maximum transverse shear load in stringer simply as:

$$V = 0.5 \cdot p \cdot B \cdot L$$

- Combined axial and bending be treated as classical beam problem, assuming clamped-guided conditions at frame supports. Thus, critical bending moments shall be:

✓ $P_x > 0$ (tension). Max. bending moment at centre $M^* = p \cdot B \cdot L^2 / 6$

, and critical bending at supports $M^* = - p \cdot B \cdot L^2 / 3$ (Ref. 9, Table 8.1)

- ✓ $P_x < 0$ (compression). Critical bending moments to be obtained as per formulas in chapter §3145 "BEAM-COLUMNS", section 2.2.1 case #12

Note: M positive if inner points of the stringer are in compression

5.5 JOINT BETWEEN STRINGER AND SKIN

The shear load that shall be sustained by the joint between skin and stringer can be calculated simply as the difference between the shear loads in skin subpanels at both sides of the stringer:

$$q = N_{xy1} - N_{xy2} \quad \dots \text{if pressure on panel:} \quad q = p \cdot B / h \quad (\text{see Figure 5})$$

If n represent the rivet rows attaching the stringer to the skin (typically 1 or 2) and p_r the distance between rivets in the longitudinal direction (typ. 4 – 6 times the rivet diameter), then the shear load per rivet would be:

$$Q = q \cdot p_r / n$$

6 STRENGTH AND STATIC MARGINS

As discussed in section §3, the response of the structure to applied loads shall be typically non-linear (specially in compression cases) and the reserve factor calculation shall consist of a numeric search scaling each set of applied loads until the first failure index, among all the relevant failure modes, is equal to 1.0. The amplification factor of applied loads at such stress state shall be the reserve factor.

6.1 STRINGER

Material strength under direct loads

- In tension, compare the longitudinal stress vs the material ultimate tension stress:

$$RF = F_{tu} / f_{cg} \quad \dots \text{or, expressed as failure index: } R = f_{cg} / F_{tu}$$

- In compression, the crippling allowable stress would be used instead. However, this failure mode is covered by the column buckling margins obtained later on (which contains the crippling cut-off as limit for short beams)

See also the discussion about the strength analysis of the super-stringer cross-section with the skin panel under pressure load in section 6.3.

6.2 SKIN

Material strength under direct loads

Compare Von-Mises stress (with direct stress components, i.e. load flows divided by skin thickness) with the material ultimate tension stress.

$$RF = F_{tu} / \sigma_{vm} \quad \dots \text{or, expressed as failure index: } R = \sigma_{vm} / F_{tu}$$

Skin shear strength

Melcon (Ref. 7) defines a dedicated allowable value for the shear stress of the skin, which shall account for the extended strength of the material developing diagonal tension. This shear allowable depends also on the longitudinal stress (then it is not a "pure allowable").

$$F_s = F_{scr} + [F_{so} - F_{scr}] / \eta$$

where:

F_{so} shall be obtained interpolating in curves of FIGURE 5 in Ref. 7 (note that for materials not included in graph the value of F_{so} at $f_s/F_{scr} = 1$ is equal to $0.75 \cdot F_{su}$)

$$\eta = \{ [1 + (a/b)^2] / 2 \}^{0.25} \quad \text{limiting the } a/b \text{ ratio within the interval between 1.0 and 3.0}$$

The dedicated allowable stress proposed for post-buckled skins in shear shall be compared with the direct shear stress in the skin:

$$RF = F_s / f_s$$

$$\dots \text{or, expressed as failure index: } R = f_s / F_s$$

Local buckling

The level of buckling (margin indicator analogue to RF) is presented only for information, as the post-buckling of skin below ultimate loads is typically permitted. Expressions to compute this margin under combined loads shall be:

- Tension ($\sigma > 0$) and shear (τ)

Failure indices

$$R_t = \sigma / F_{ccr}$$

$$R_s = \tau / F_{scr}$$

Bruhn C11.32

$$R_s - 0.5 \cdot R_t = 1$$

$$LB = 1 / (R_s - 0.5 \cdot R_t)$$

- Compression ($\sigma < 0$) and shear (τ)

Failure indices

$$R_c = -\sigma / F_{ccr}$$

$$R_s = \tau / F_{scr}$$

Bruhn C11.32, eq. C5.12

$$R_c + R_s^2 = 1$$

$$LB = 2 / [R_c + (R_c^2 + 4 \cdot R_s^2)^{0.5}]$$

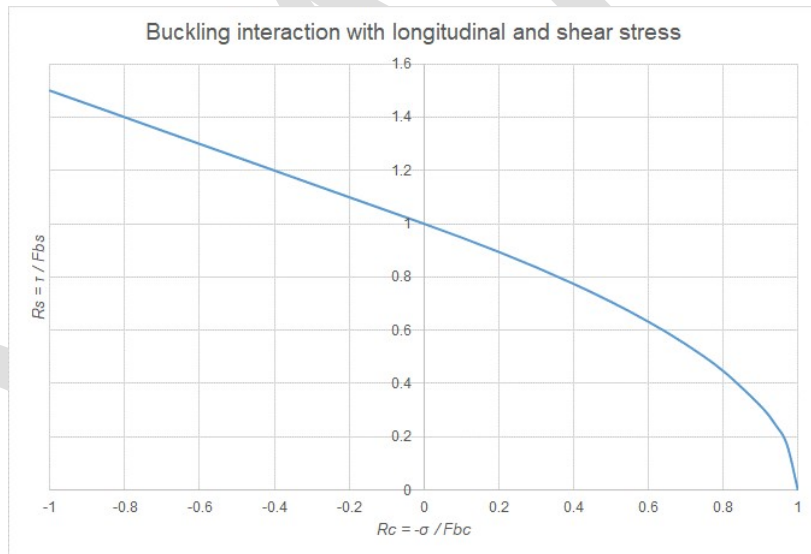


Figure 6. Buckling interaction with longitudinal and shear stress

6.3 SUPER-STRINGER

Column buckling in compression

Direct compression load should not exceed the super-stringer column buckling load (crippling cut-off accounted). The associated failure index:

$$R_1 = f_c / F_c$$

(can be calculated based in forces instead of stresses)

Global buckling in shear

On the other hand, shear load in skin beyond buckling onset will induce compression stresses in stringer (diagonal tension). Additionally, the fictitious stress computed for monitoring the effective support provided by the stringer to the skin shall be added for evaluating the failure index in this case:

$$R_2 = (f'_c + f''_c) / F'_c$$

Interaction equation

Finally, the interaction law proposed by Melcon to account for combined effects of compression and shear in the stiffened pane, at failure loads level, is:

$$R_{1-u}^{1.25} + R_{2-u}^{1.25} = 1$$

At any other level of applied loads:

$$R = R_1^{1.25} + R_2^{1.25} \quad \text{or, for better consistency} \quad R = (R_1^{1.25} + R_2^{1.25})^{1/1.25}$$

Cross-section strength with pressure load on panel

If the stiffened panel is subjected to pressure loads, use the combined shear, axial and bending forces calculated as shown in section §5.4 to get the stress distribution in the section (see chapter §3110 "BEAMS-SECTIONS") to get the static margins at the critical cross section.

Riveted joint

The static margin of the riveted joint between the stringer and the skin shall be calculated using skin material and fastener allowables, by-pass loads and rivet loads discussed in this document previously, and method in chapters §3220 "FASTENERS AND BOLTS FAILURE" and §3230 "PLATE FAILURE: BEARING AND PULL-THROUGH".

7 REMOVABLE SKIN PANELS

Some fuselage skin panels may be designed to permit a simple removal for facilitating internal access and maintenance activities on ground. These panels would typically not be continuous across stringers and/or frames supports. Additionally, the attachment of the panel to the stiffening member would typically be done with fastening systems allowing simple disassembly (typ. bolt & nut), installed with medium-to-high clearance between the bolt and the hole.

The suggested approach to deal with these configurations in analysis shall consist in using the previously describe method with the following considerations:

- Assume the removable panel will not take the longitudinal X loads (then, sustaining only in-plane shear and transverse hoop load). All the longitudinal X load that this panel would have to sustain shall be transferred to the stringer (and eventually the adjacent skin panel if this other can be considered a working panel)
- Assume the removable panel will not contribute with any effective skin width to the column stability of the stiffened panel. In case that, at one analysis location (the super-stringer), both panels are non-working panels, the whole axial load shall have to be sustained only by the stringer (anyway the column buckling strength evaluated only regarding bending instability across the axis parallel to the skin)
- Dedicated analysis of the riveted joints shall be needed if panels are not continuous across stiffening members, as in that case these fastener are directly responsible of transferring the axial (longitudinal and/or transverse) and shear loads at the perimeter of the panels.

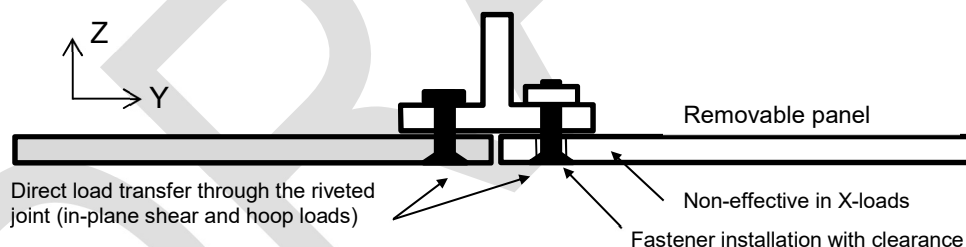


Figure 7. Sketch of cross section at location with a removable panels

8 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
E	MPa	Elastic modulus of the material
E_c	MPa	Compression modulus of the material
t	mm	Thickness of the skin panel
a, L	mm	Length of the skin panel (distance between frames supports)
b	mm	Width of the skin panel (arc distance between stringers)
R	mm / -	Curvature radius of the skin panel, or failure index for a certain type of load and failure mode (indicated by the subscript)
p_r	mm	Rivets pitch, i.e. distance between rivets, in the longitudinal direction, in this case
I_{st}	mm ⁴	Moment of inertia of stiffener about centroidal axis parallel to the tangent of the skin contour
J_{st}	mm ⁴	Torsional stiffness factor. For open thin walled sections equal to $\sum(b_i \cdot t_i^3) / 3$
f_s, τ	MPa	Direct shear stress in skin
P_x	N	Total axial-X load force in the super-stringer element (stringer plus halves of adjacent skin panels)
N_y	N/mm	Transverse-Y load flow in skin panels
N_{xy}	N/mm	Shear in-plane load flow in skin panels
p	MPa	Pressure on skin (inner pressure positive)
f_{cg}	MPa	Longitudinal/axial stress in the section, normalized with the total super-stringer cross section area
f_c	MPa	Longitudinal/axial stress in the section, normalized with the effective area of super-stringer cross section (appropriate effective skin width in compression with the post-buckled skin)
σ	MPa	Longitudinal/axial stress
f_s	MPa	Direct shear stress in skin
q	N/mm	Shear load flow to be sustained by the riveted joint between stringer and skin
Q	N	Shear load force at each rivet of the joint between stringer and skin
F'_c	MPa	Column buckling stress of the stringer
F_c	MPa	Column buckling stress of the super-stringer assembly (stringer plus skin)
F_s	MPa	Allowable shear stress of the skin considering post-buckling
F_{ccr}	MPa	Local buckling stress of the skin in compression
F_{scr}	MPa	Local buckling stress of the skin in shear
F_{tu}	MPa	Ultimate tensile stress of the material

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[Michael C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	[S.P. Timoshenko y J.M. Gere]	2 nd Ed.	Theory of Elastic Stability
Ref. 4	NACA TN 3781	1957	Handbook of Structural Stability. Part I.- Buckling of Flat Plates
Ref. 5	NACA TN 3783	1957	Handbook of Structural Stability. Part III.- Buckling of Curved Plates and Shells
Ref. 6	NACA TN 2661	1952	A Summary of Diagonal Tension. Part I.- Methods of Analysis
Ref. 7	[M.A. Melcon y A.F. Ensrud]	1953	Analysis of Stiffened Curved Panels Under Shear and Compression
Ref. 8	NT-T-ADP-98009	Ed. C	Manual Teórico del Programa CRLARDP, versión 3.x
Ref. 9	[Warren C. Young]	7 th Ed.	Roark's Formulas for Stress and Strain

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
CRLARDP	3.6	

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.5 27/05/2022	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños 27/05/2022	Florian Glock DD/MM/AAAA

Table 3. Approval signatures table for last revision of this chapter

DRAFT